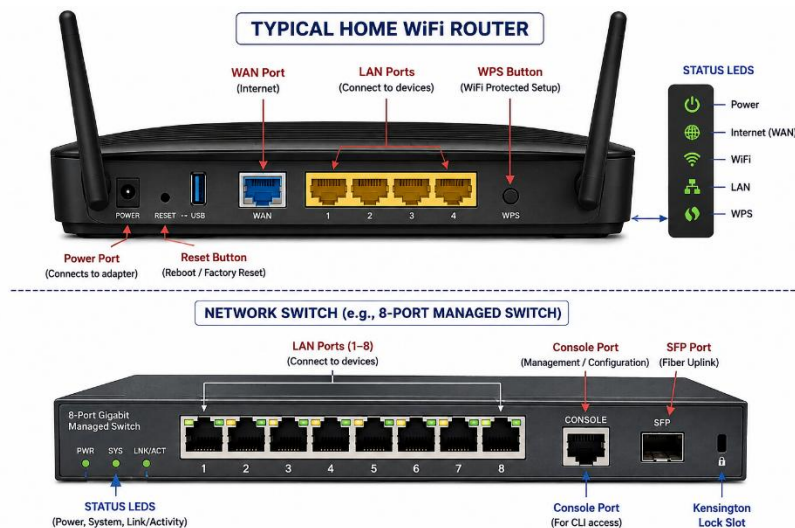


ASSIGNMENT:2

Routers and Switches Theory

- ❖ Task (1): Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs.

Ans.



- ❖ **Task (2): List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic. Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.**

Ans.

- **Differences Between Routers and Switches:**

	Router	Switch
1. Main Function	Connects different networks, such as a home network to the Internet.	Connects devices within the same local network (LAN).
2. Addresses Used	Uses IP addresses to decide where to send packets.	Uses MAC addresses to forward Ethernet frames to the correct device.
3. Traffic Handling	Routes traffic between different networks and can connect a private LAN to the Internet/WAN.	Forwards traffic within the same LAN , usually sending it only to the intended device's port.

❖ **Task (3): Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.**

Ans.

❖ **Router and Network Switch – Port Identification:**

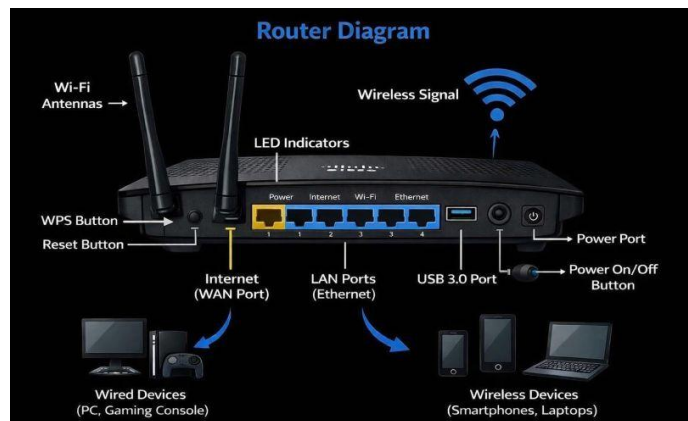
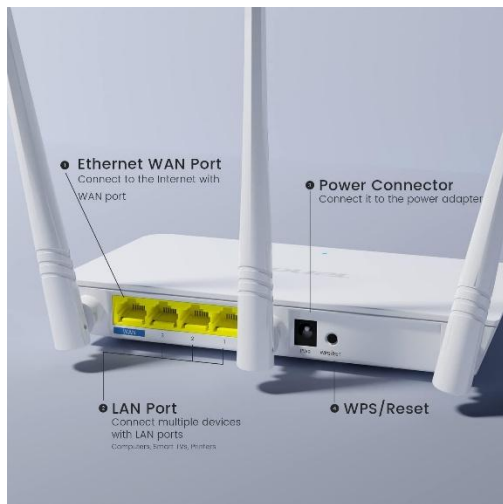
1. Wi-Fi Router:

- **Power Port:** Located at the back of the router. It connects the router to the power adapter and supplies electrical power.
- **WAN/Internet Port:** Usually located at the back and labelled WAN or Internet. It connects the router to the modem or Internet service.
- **LAN Ethernet Port 1:** Used to connect a computer or other wired device to the local network.
- **LAN Ethernet Port 2:** Used to connect another wired device, such as a printer or desktop computer.
- **LED Indicators:** Usually located on the front of the router. They indicate Power, Internet/WAN, Wi-Fi, and LAN activity/status.

2. Network Switch:

- **Power Port:** Usually located at the back of the switch. It connects to the power adapter or power supply.
- **Ethernet Port 1:** Used to connect one computer or network device to the switch.
- **Ethernet Port 2:** Used to connect another computer or network device to the same local network.
- **LED Indicators:** Located near the Ethernet ports or on the front panel. They show Power, Link, and Network Activity.





- ❖ **Task (4):** Imagine you are setting up a small gaming cafe network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.

Ans.

Network setup:

- ISP → Router → Switch → Gaming PCs

- I am setting up a small gaming cafe for IPL streaming and multiplayer gaming. I would first connect the ISP Internet line to the router's WAN/Internet port. The router will provide Internet access and manage the network using IP addresses.

- Next, I would connect a network switch to one of the router's LAN ports using an Ethernet cable. I would then connect all 8 gaming PCs to the switch using separate Ethernet cables.
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